

The programmable model of 8086 through the core 2:

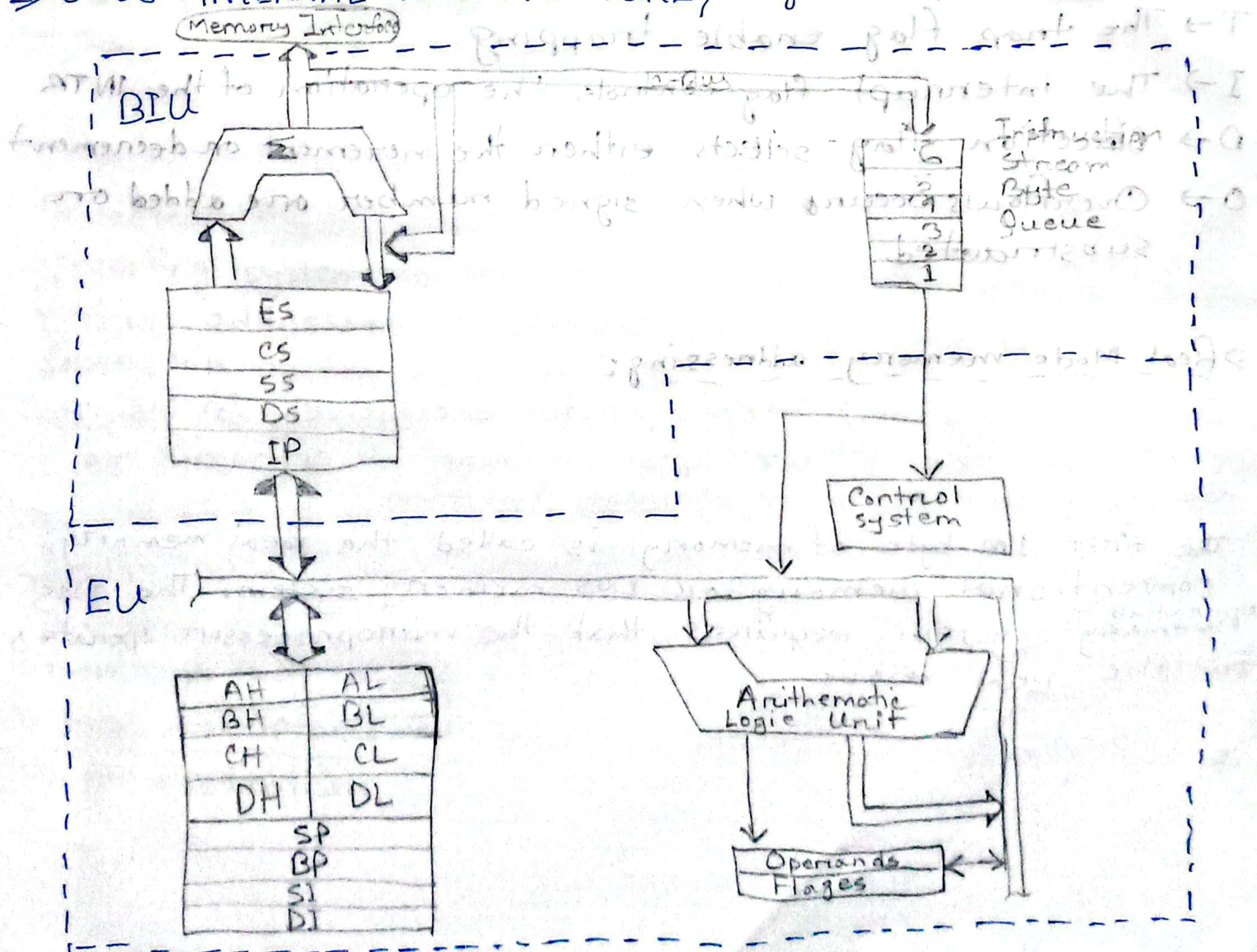
Register size Overlaid

16-bit	32-bit	64-bit
RAX	EAX	
RBX	EBX	
RCX	ECX	
RDX	EDX	
RBP	EBP	
RSP	ESP	
RDI	EDI	

⇒ Typical working relationships:

① AX ↔ BP/SP ② AX/BX/CS ↔ SI/DI ③ CX ↔ SI/DI ④ AX ↔ DX

⇒ 8086 INTERNAL ARCHITECTURE/Organization



→ Flag Register of 8086 :



- C → carry flag holds the carry after addition or the borrow
- P → Parity is a logic 0 for odd parity and a logic 1 for even parity.
- A → Auxiliary carry holds the carry after addition or borrow after subtraction between bit positions 3 and 4 of the result.
- Z → zero flag shows the result of an arithmetic if $Z=1$ the result is zero and $Z=0$ the result is not zero.
- S → The sign flag --- If $S=1$ the sign bit is negative if $S=0$ then positive
- T → The trap flag enable trapping
- I → The interrupt flag controls the operation of the INTR
- D → Direction flag selects either the increment or decrement
- O → Overflows occurs when signed number are added or subtracted.

⇒ Real Mode memory addressing:

Only the 8086 and 8088 operate exclusively in the real mode. In the 64-bit operation mode of pentium 4 and core 2 there is no real mode operation.

The first 1M byte of memory is called the real memory. conventional memory or DOS memory system. The DOS operating system requires that the microprocessor operate in the real mode.

⇒ Segment & Offsets:

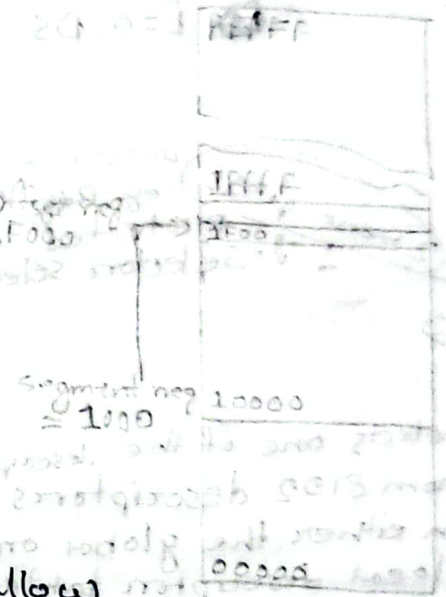
an offset address chooses a location within a segment in real mode.

The segment address, located within one of the segment registers, defines the beginning address of any 64 K-byte memory segment.

The offset address selects any location within 64 K-byte memory segment.

⇒ Default 16-bit segment and offset combinations:

Segment of 16-bit	Offset	Segment Purpose
CS	IP	Instruction add.
SS	SP, BP, SI, DI	Stack add.
DS	BX, DI, SI, 8/16-bit	Data add.
ES	DI for string inst.	String destination add.



⇒ Protected Mode Memory Addressing:

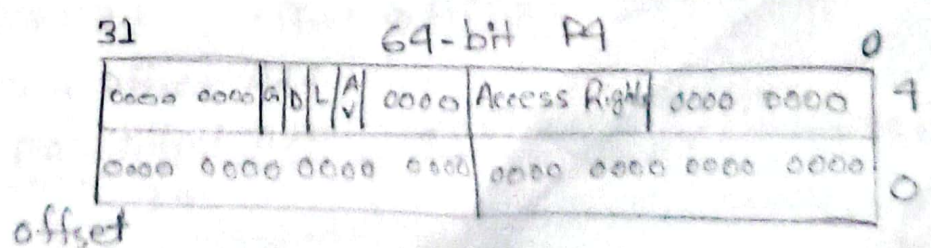
Protected mode memory addressing allow operating system to control and protect memory by mapping virtual addresses to physical memory using selectors, descriptor tables and paging.

selector: The selector, located in the segment register, selects one of the 8192 descriptors from one of two tables of descriptors.

Descriptor: The descriptor describes the location, length, and access rights of the segment of memory.

↳ Global descriptor: The global descriptors contains segment definitions that apply to all programs.

↳ Local descriptor: The local descriptors are usually unique to an application.



⇒ Paging Register:- A paging register is a CPU control register such as the CR3 register that holds the base address for the multi-level tables used in memory paging.

⇒ Step of execute a machine instruction:

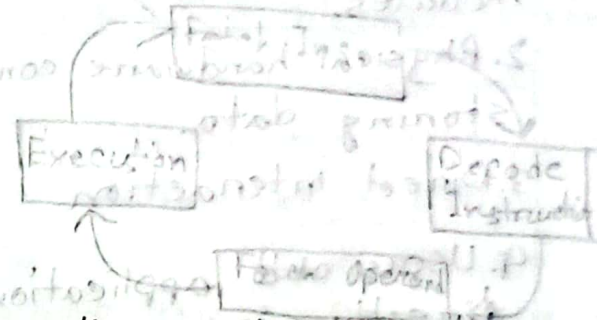
Machine instruction are execute in 4 steps which are listed as follows:

① Fetch instruction: Done by BIU

② Decode instruction: Done by EU

③ Fetch operands: Done by BIU

④ Execution: Done by EU.



Both Bus interface unit (BIU) and Execution Unit (EU) works mutually exclusive to each other to execute the machine instruction.

⇒ Define memory segment:

Memory Segment:

Memory segment is the process in which the main memory of the computer is logically divided into different segment and each segment has its own base address.

⇒ The feature of 80286 microprocessor:

① 80286 is a high performance 16-bit microprocessor.

② 80286 base address is a 24 bit address, so it could address upto 16 MB of physical memory.

③ 80286 supports protected mode memory addressing.

④ Multitasking ability

⑤ 80286 has improved bus interface.

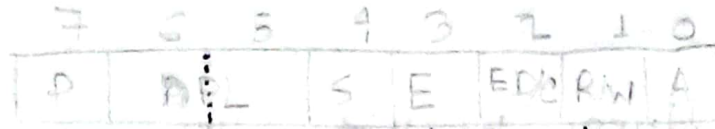
Math:

⇒ Physical address :- 80FD2h ; offset ⇒ BFD2h ; seg = ?

PA = segment Add. x 10 + offset Add.

$$\text{segment Add.} = \frac{\text{PA} - \text{offset Add.}}{10} = \frac{80FD2h - BFD2h}{10} = 7500h$$

⇒ The access right byte for the 80286: through ~~Port 2~~ descriptor:



(Descriptor is undefined) $P=0$
 (Seg. contain a valid base and limit) $P=1$

DLP = sets the descriptor privilege level

System descriptor, $S=0$, $S=1$, CS/DS descriptor

$E=0$, DS / $E=1$, CS

segment not access
 segment has access

→ $R=0$, CS may not be read
 $R=1$, CS may be read
 $W=0$, DS may not read
 $W=1$, DS may read

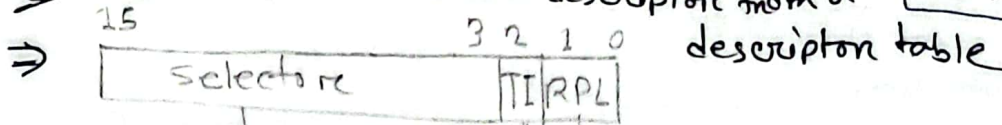
→ $C=0$, Ignore descriptor privilege level

→ $C=1$, Abide by Privilege level

→ $ED=0$, expand upward (DS)

→ $ED=1$, expand downward (CS)

Describe the content of seg. Reg.
 seg. Reg. Contains a Selector of protected mode memory address
 18. Selector select a descriptor from a



selects one of the descriptors from 8192 descriptors in either the global or local descriptor table.

RPL = Requested privilege level where 00 is the highest and 11 is lowest

TI = 0, Global descriptor

TI = 1, Local descriptor

⇒ Program-Invisible Registers:

The program-invisible register is often called cache memory because cache ~~are~~ is any memory that store information. This cache is not to be confused with the level-1 or level-2 caches found with the microprocessors.

⇒ Difference between physical and logical memory:

Physical Memory	Logical Memory
1. Physical memory refers to the actual hardware components where data and instructions are stored in a computer system. It consists of RAM modules.	1. Logical memory is the virtual address space that a program or process uses during execution.
2. Physical hardware components storing data.	2. Virtual representation of available memory.
3. Direct interaction by CPU	3. Accessed through logical addresses.
4. Users and applications don't directly interact.	4. Program and users work with logical address.
5. Managed by hardware and operating system.	5. Primarily managed by the operating system.

⇒ The process of the local descriptor table addressed in the main memory system?

The location of the local description table is selected from the global descriptor table. One of the global descriptors is set up to address the local description table. To access the local descriptor table, the LDTR (Local Descriptor Table Register) is loaded with a selector, just as a segment register loaded with a selector. This selector accesses the global descriptor table and loads the address, limit, access right of the local descriptor table into the cache portion of the LDTR.

⇒ Which Intel microprocessor address 1T of memory?
Pentium 4 and core 2 Intel microprocessor address 1T of memory.

⇒ The GDTR (Global descriptor Table Register) register locates the global descriptor table.

⇒ What is the purpose of the microprocessor in a microprocessor based computer?

Microprocessors contain the arithmetic, logical and control circuit required to perform the function of a CPU. It is responsible for processing the unique set of instructions and process that make up a computer program.

The microprocessor is a multipurpose digital integrated circuit that accepts binary data as input process it according to instructions stored in its memory and provides results as outputs.

⇒ Program-visible registers?

Program visible registers are the registers that are directly used in an instruction.

The programming model of the 8086 through the pentium 4 is ~~also~~ considered to be program visible because its registers satisfy above condition.

⇒ Protected mode memory addressing allows access to which area of the memory in the 80286 —.

Protected mode memory addressing allows access to data and program located above the first 1M byte of memory as well as within the first 1M byte of memory.

3 Determine the memory location addressed by the following real mode code2 register combination.

① $\text{DS} = 2000\text{H}$ and $\text{EAX} = 00003000\text{H}$

$$\begin{aligned}\text{memory location} &= 2000\text{H} \times 10\text{H} + 3000\text{H} \\ &= 23000\text{H}\end{aligned}$$

⇒ Which registers are used as an offset address for the string instruction destination in the microprocessor?

The Destination Index (DI) register is used as an offset register address for the string instruction destination in the microprocessor.

⇒ Find the memory add. of the next instruction executed by the μP . When operated in the real mode, for the following $\text{CS}:\text{IP}$ combinations.

① $\text{CS} = 1000\text{H}$ and $\text{IP} = 2000\text{H}$

$$\text{PA} = 1000 \times 10\text{H} + 2000\text{H} = 12000\text{H}$$

② $\text{CS} = 3956\text{H}$ $\text{IP} = \text{ABCDH}$: 12000H

$$\begin{aligned}\text{PA} &= 3956 \times 10\text{H} + \text{ABCDH} \\ &= 3\text{F}12\text{D H}\end{aligned}$$

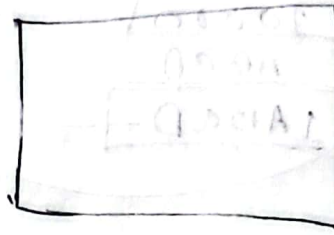
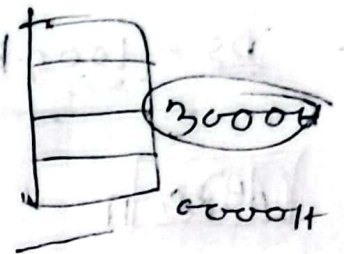
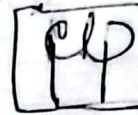
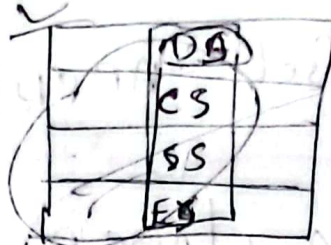
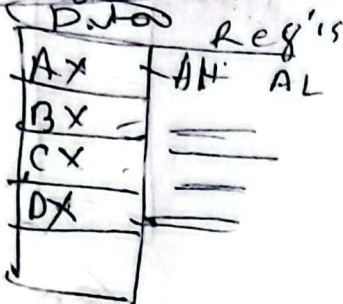
∴ Next instruction memory address: $3\text{F}12\text{D H}$

⇒ Purpose of IP/EIP register?

Hold the offset address of the next step in the program. The IP/EIP register is used to store the IP address of the next instruction. The instruction pointer, which points to the next instruction in a program, is used by the μP to find the next sequential instruction in a program located within the code segment.

7009
8008
8086

Data register



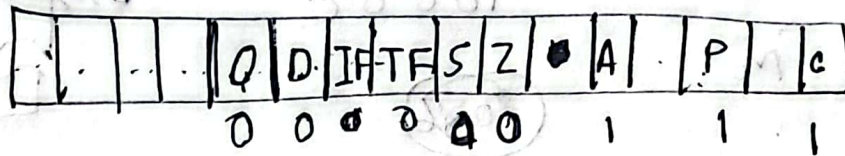
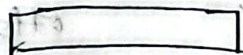
Pointer & Indexing -

BP



SI
DI

void to f



2
2

8086

$$\begin{array}{r} 11001011 \\ 10101011 \\ \hline 101110110 \end{array}$$

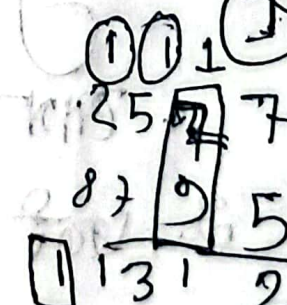
8

8x8

64

110

110



$$\begin{array}{r} 11001011 \\ 10101011 \\ \hline \end{array}$$

1100 1011
1010 1011

1

2527

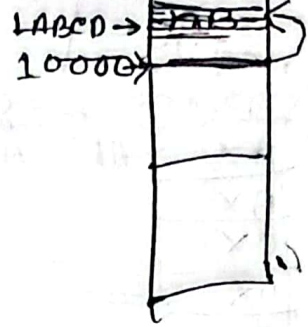
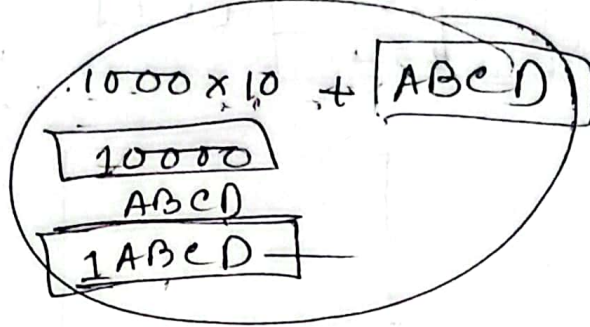
8325

2

$$\begin{array}{r} 0111 \\ 0101 \\ \hline \end{array}$$

DS - 1000H offset - ABCDH

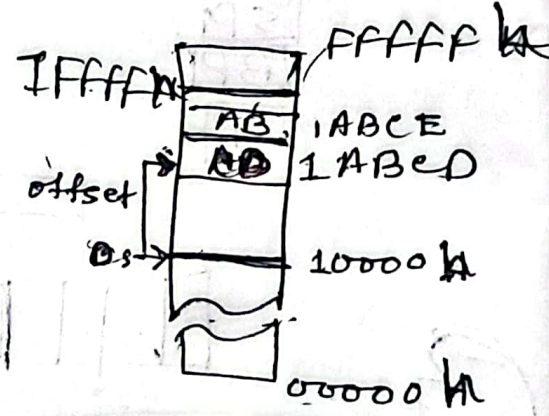
AB



$$\begin{array}{r} \text{ABCD} \\ \text{FFFF} \\ \hline 10000H \\ \text{FFFF} \\ \hline 1FFFF \end{array}$$

ABCD

8086
Phy / D / M



feature

2²⁰
10000

- 4. Protected mode
- 5. Flat mode 8086 or 8088

1. Register এর পরিচয় intel-2
2. Flag কমান্ডে ব্যবহৃত visible invisible
সহজ বিজ্ঞান গ্রন্থ সহজ
3. একটি DS এর offset সহজ আকারে
আমাদের memory Representation
সহজ হলে DATA সহজ বসে
হলে সহজ করে